Date: ___/___/___

What is ~~the~~ TF-IDF (intuition)

$$TF = \frac{\text{Number of times a word 'x' appears in a Document}}{\text{Number of word present in a Document}}$$

$$IDF = \log\left(\frac{\text{Number of } \cancel{\text{times a word 'x' appears in}} \text{ Document Present in a corpus}}{\text{Number of Documents where word 'x' has appeared}}\right)$$

$$\boxed{TF\,IDF = TF * IDF}$$

Document	Text
D1	students watch youtube
D2	youtube watch youtube
D3	student write comment
D4	youtube write comments

Term-frequency:

D1 = students watch youtube

No of word = 3

	TF
student	1/3
watch	1/3
youtube	1/3

D2 = youtube watch youtube

No of word = 3

	TF
youtube	2/3
watch	1/3

* IDF এ যে word বেশি Rare তার IDF বেশি।

IDF = 4

IDF = 4

	IDF	
student	$\log(4/2)$	→ 0.69
watch	$\log(4/2)$	→ 0.69
youtube	$\log(4/3)$	→ 0.28
comment	$\log(4/2)$	→ 0.69
write	$\log(4/2)$	→ 0.69

D1, D2, D3, D4 সব মিলে [illegible]

* TF → Doc এ বেশি বার আসবে তার value বেশি।

TF·IDF = TF * IDF

for watch:

$= 1/3 * \cancel{0.28}\ 0.69$

$= 0.27$